

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-06

Appendix-F

Non-Functional Requirement Catalogue

School ERP

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Classification: Confidential — Client & Project Team Only

About This Appendix

This document is Appendix-F to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-06: Non-Functional Requirement Catalogue. It defines every system quality attribute required before Architecture Design and SRS, across all 40 requested NFR categories.

This is a new, standalone document. RGD v2.0, Appendix-A, Appendix-B, Appendix-C v1.1, Appendix-D, and Appendix-E are not modified, opened for editing, or altered in producing this document — all cross-references (Related FRs, Related Business Rules) point back to those documents by ID only, using the same mappings already established there.

Total NFRs catalogued: 88, across all 40 requested categories. Where a target value or dependency is not yet confirmed by the client, it is marked "Client/Product Decision Required" within the relevant field rather than assumed.

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Detailed Non-Functional Requirements

Category: Performance (12 NFRs)**NFR-PERF-001 — Page Load Time**

1. NFR ID	NFR-PERF-001
2. Title	Page Load Time
3. Category	Performance
4. Description	General page load time across the application under normal load.
5. Business Objective	Ensure a responsive user experience for all roles.
6. Rationale	Slow page loads directly reduce staff productivity and parent portal adoption (RGD v2.0 Success Criteria).
7. Acceptance Criteria	95% of page loads complete within target under normal load, verified by load test report.
8. Measurement Method	Load testing (95th percentile response time)
9. Target Value	≤3 seconds for 95% of requests
10. Priority	High
11. Risk	Medium
12. Dependencies	Network Security (NFR-NET-001), Caching (NFR-CACHE-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-002 — Login Response Time

1. NFR ID	NFR-PERF-002
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2. Title	Login Response Time
3. Category	Performance
4. Description	Time from credential submission to authenticated dashboard render.
5. Business Objective	First-impression responsiveness; reduces front-office login-related complaints.
6. Rationale	A slow login is the single most visible performance metric to every daily user.
7. Acceptance Criteria	95% of logins complete within target, verified by load test.
8. Measurement Method	Load testing
9. Target Value	≤2 seconds
10. Priority	High
11. Risk	Medium
12. Dependencies	Session Management (NFR-SESS-001), Authentication (NFR-AUTHN-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	NFR-AUTHN-001, NFR-SESS-001
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-003 — Dashboard Loading

1. NFR ID	NFR-PERF-003
2. Title	Dashboard Loading
3. Category	Performance
4. Description	Time to render all role-based widgets on the Reports & Analytics dashboard.

5. Business Objective	Support real-time operational decision-making (RGD v2.0 OBJ).
6. Rationale	A slow dashboard undermines the value proposition of consolidated reporting.
7. Acceptance Criteria	95% of dashboard loads complete within target for a role with the maximum widget count.
8. Measurement Method	Load testing
9. Target Value	≤3 seconds
10. Priority	High
11. Risk	Medium
12. Dependencies	Caching (NFR-CACHE-001), Database Performance (NFR-DBP-001)
13. Applicable Modules	Reports & Analytics
14. Related FRs (Appendix-E)	FR-40
15. Related Business Rules (Appendix-C)	BR-RPT-004
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-004 — Search Response

1. NFR ID	NFR-PERF-004
2. Title	Search Response
3. Category	Performance
4. Description	Response time for student/staff/library catalog search queries.
5. Business Objective	Support efficient daily operations for Admin Staff and Librarian.
6. Rationale	Search is a high-frequency interaction; delays compound across many daily searches.
7. Acceptance Criteria	95% of search queries return within target at current data scale.

8. Measurement Method	Load testing
9. Target Value	≤2 seconds
10. Priority	Medium
11. Risk	Low
12. Dependencies	Search Performance (NFR-SRCH-001)
13. Applicable Modules	SIS, Library
14. Related FRs (Appendix-E)	FR-06, FR-27, FR-29
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-005 — Report Generation

1. NFR ID	NFR-PERF-005
2. Title	Report Generation
3. Category	Performance
4. Description	Time to generate standard, pre-built reports.
5. Business Objective	Ensure management/finance reports are available on demand.
6. Rationale	Delayed reports reduce the value of the Reports & Analytics module.
7. Acceptance Criteria	95% of standard reports generate within target; larger custom reports follow NFR-RPTP-001.
8. Measurement Method	Load testing
9. Target Value	≤10 seconds for standard reports
10. Priority	Medium

11. Risk	Medium
12. Dependencies	Reporting Performance (NFR-RPTP-001)
13. Applicable Modules	Reports & Analytics, Fee Collection
14. Related FRs (Appendix-E)	FR-26, FR-40, FR-41
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-006 — Attendance Save

1. NFR ID	NFR-PERF-006
2. Title	Attendance Save
3. Category	Performance
4. Description	Time to save a single attendance-marking action.
5. Business Objective	Attendance marking happens many times daily per teacher; must feel instantaneous.
6. Rationale	Slow saves directly reduce teacher willingness to use the digital system over paper.
7. Acceptance Criteria	95% of save actions complete within target.
8. Measurement Method	Load testing
9. Target Value	≤1 second per save action
10. Priority	High
11. Risk	Low
12. Dependencies	Database Performance (NFR-DBP-001)
13. Applicable Modules	Attendance Management

14. Related FRs (Appendix-E)	FR-10
15. Related Business Rules (Appendix-C)	BR-ATT-001
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-007 — Fee Payment Processing

1. NFR ID	NFR-PERF-007
2. Title	Fee Payment Processing
3. Category	Performance
4. Description	Time from payment gateway confirmation to ledger update and receipt availability.
5. Business Objective	Give parents immediate payment confirmation, reducing support queries.
6. Rationale	A delayed receipt after a successful payment causes parent anxiety and duplicate-payment attempts.
7. Acceptance Criteria	95% of confirmed payments update the ledger and issue a receipt within target.
8. Measurement Method	Load testing
9. Target Value	≤5 seconds
10. Priority	High
11. Risk	High
12. Dependencies	<i>Integration Dependency: Payment Gateway (Client/Product Decision Required)</i>
13. Applicable Modules	Fee Collection
14. Related FRs (Appendix-E)	FR-23, FR-24
15. Related Business Rules (Appendix-C)	BR-FEE-001, BR-FEE-006
16. Related Security Controls	NFR-NET-002

17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-PERF-008 — Bulk Import

1. NFR ID	NFR-PERF-008
2. Title	Bulk Import
3. Category	Performance
4. Description	Processing time for bulk student/staff data import.
5. Business Objective	Support efficient onboarding at go-live and year-start.
6. Rationale	Large imports (e.g., 300+ students) must complete in a practical operational window.
7. Acceptance Criteria	1,000 records process within target under standard conditions.
8. Measurement Method	Load testing with representative data volume
9. Target Value	≤2 minutes per 1,000 records
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Database Performance (NFR-DBP-001)
13. Applicable Modules	SIS, HR
14. Related FRs (Appendix-E)	FR-07
15. Related Business Rules (Appendix-C)	BR-SIS-003
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency

20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-009 — Export

1. NFR ID	NFR-PERF-009
2. Title	Export
3. Category	Performance
4. Description	Time to export report/list data to Excel/PDF.
5. Business Objective	Support offline analysis and record-keeping needs.
6. Rationale	Slow exports interrupt user workflow, especially for Finance Team and Academic Head.
7. Acceptance Criteria	Exports up to 10,000 rows complete within target; larger exports follow NFR-RPTP-001.
8. Measurement Method	Load testing
9. Target Value	≤15 seconds up to 10,000 rows
10. Priority	Medium
11. Risk	Low
12. Dependencies	Reporting Performance (NFR-RPTP-001)
13. Applicable Modules	Reports & Analytics, Fee Collection
14. Related FRs (Appendix-E)	FR-26, FR-41
15. Related Business Rules (Appendix-C)	BR-RPT-001
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-010 — Notification Delivery

1. NFR ID	NFR-PERF-010
2. Title	Notification Delivery
3. Category	Performance
4. Description	Time from trigger event to dispatch at the Communication gateway.
5. Business Objective	Ensure notifications (absence, fee reminders) are timely enough to be actionable.
6. Rationale	A delayed absence notification defeats its safeguarding purpose.
7. Acceptance Criteria	95% of triggered notifications dispatch to the gateway within target.
8. Measurement Method	Load testing plus gateway-latency monitoring
9. Target Value	≤60 seconds
10. Priority	High
11. Risk	High
12. Dependencies	<i>Integration Dependency: SMS/Email Gateway (Client/Product Decision Required)</i>
13. Applicable Modules	Communication & Notifications
14. Related FRs (Appendix-E)	FR-05, FR-11, FR-25, FR-37
15. Related Business Rules (Appendix-C)	BR-COM-002, BR-COM-003
16. Related Security Controls	NFR-NET-002
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-PERF-011 — API Response Time

1. NFR ID	NFR-PERF-011
2. Title	API Response Time

3. Category	Performance
4. Description	Response time for internal application API calls (not third-party gateways).
5. Business Objective	Ensure the frontend/backend architecture itself is not a bottleneck.
6. Rationale	API latency compounds across a page that makes multiple calls.
7. Acceptance Criteria	95th percentile of internal API calls complete within target.
8. Measurement Method	Load testing (API-level)
9. Target Value	≤500 ms
10. Priority	High
11. Risk	Medium
12. Dependencies	API Performance (NFR-APIP-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-PERF-012 — Database Query Response

1. NFR ID	NFR-PERF-012
2. Title	Database Query Response
3. Category	Performance
4. Description	Response time for indexed database queries under production-scale data.
5. Business Objective	Ensure database design does not degrade as data volume grows over academic years.

6. Rationale	Unindexed or poorly optimized queries are the most common cause of ERP performance decay over time.
7. Acceptance Criteria	95th percentile of indexed queries complete within target at maximum scalability targets (NFR-SCAL series).
8. Measurement Method	Query-plan analysis under load
9. Target Value	≤200 ms
10. Priority	High
11. Risk	High
12. Dependencies	Database Performance (NFR-DBP-001), Scalability series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load testing (jmeter/k6) under simulated peak concurrency
18. Validation Method	Automated performance test suite run pre-release
19. Verification Method	Load testing (JMeter/k6) under simulated peak concurrency
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Scalability (8 NFRs)**NFR-SCAL-001 — Concurrent Users**

1. NFR ID	NFR-SCAL-001
2. Title	Concurrent Users
3. Category	Scalability
4. Description	Maximum simultaneous active users the system must support without performance degradation.
5. Business Objective	Support peak usage (e.g., morning attendance window, fee-due-date payment surge).
6. Rationale	RGD v2.0 NFR assumed this figure; it remains unconfirmed pending client enrollment/staff count data.
7. Acceptance Criteria	System sustains target concurrency with Performance NFRs still met.
8. Measurement Method	Load/stress testing
9. Target Value	<i>2,000 concurrent users (Client/Product Decision Required to confirm against actual enrollment)</i>
10. Priority	High
11. Risk	High
12. Dependencies	Performance series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-SCAL-002 — Maximum Students

1. NFR ID	NFR-SCAL-002
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2. Title	Maximum Students
3. Category	Scalability
4. Description	Maximum active student records the system must support per instance.
5. Business Objective	Ensure the platform scales with enrollment growth, including multi-campus scenarios (RGD v2.0 OBJ-05).
6. Rationale	Undersized capacity planning risks a costly re-architecture as the school grows.
7. Acceptance Criteria	System performs within Performance NFR targets at maximum student volume.
8. Measurement Method	Capacity/load testing at target scale
9. Target Value	<i>20,000 active students (Client/Product Decision Required to confirm)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Database Performance
13. Applicable Modules	Student Information Management
14. Related FRs (Appendix-E)	FR-06
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-SCAL-003 — Maximum Staff

1. NFR ID	NFR-SCAL-003
2. Title	Maximum Staff
3. Category	Scalability
4. Description	Maximum active staff records the system must support.
5. Business Objective	Support multi-campus HR & Payroll operation.

6. Rationale	Payroll batch performance is sensitive to total staff count.
7. Acceptance Criteria	System performs within Performance NFR targets at maximum staff volume.
8. Measurement Method	Capacity/load testing
9. Target Value	<i>2,000 active staff (Client/Product Decision Required to confirm)</i>
10. Priority	Medium
11. Risk	Low
12. Dependencies	Database Performance
13. Applicable Modules	HR & Payroll
14. Related FRs (Appendix-E)	FR-33
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-SCAL-004 — Maximum Parents/Guardians

1. NFR ID	NFR-SCAL-004
2. Title	Maximum Parents/Guardians
3. Category	Scalability
4. Description	Maximum guardian/parent portal accounts the system must support.
5. Business Objective	Ensure portal scalability given multiple guardians per student are possible.
6. Rationale	Guardian accounts can exceed student count 1.5x, affecting authentication load.
7. Acceptance Criteria	System performs within Performance NFR targets at maximum guardian account volume.
8. Measurement Method	Capacity/load testing
9. Target Value	<i>30,000 guardian accounts (Client/Product Decision Required to confirm)</i>

10. Priority	Medium
11. Risk	Low
12. Dependencies	Authentication series
13. Applicable Modules	Student Information Management, Communication
14. Related FRs (Appendix-E)	FR-06, FR-37
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-SCAL-005 — Maximum Attendance Records

1. NFR ID	NFR-SCAL-005
2. Title	Maximum Attendance Records
3. Category	Scalability
4. Description	Maximum attendance records the system must store and query per academic year without degradation.
5. Business Objective	Attendance is the single highest-volume transactional data type (multiple records per student per day).
6. Rationale	At 20,000 students × ~8 periods × ~220 days, annual volume can exceed 35 million records.
7. Acceptance Criteria	Query performance for attendance reporting (FR-13) remains within Performance NFR targets at this volume.
8. Measurement Method	Load testing with synthetic data at target volume
9. Target Value	10 million+ records per academic year
10. Priority	High
11. Risk	High
12. Dependencies	Database Performance, Archiving (NFR-ARC-001)
13. Applicable Modules	Attendance Management

14. Related FRs (Appendix-E)	FR-10, FR-13
15. Related Business Rules (Appendix-C)	BR-ATT-001
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-SCAL-006 — Maximum Exam Records

1. NFR ID	NFR-SCAL-006
2. Title	Maximum Exam Records
3. Category	Scalability
4. Description	Maximum marks/result records the system must store and query per academic year.
5. Business Objective	Support multi-exam-cycle result processing without degradation.
6. Rationale	Each student generates multiple mark records per subject per exam cycle across the year.
7. Acceptance Criteria	Grade/rank calculation (FR-19) completes within Performance NFR targets at this volume.
8. Measurement Method	Load testing with synthetic data
9. Target Value	5 million+ records per academic year
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Database Performance
13. Applicable Modules	Examination & Result Processing
14. Related FRs (Appendix-E)	FR-18, FR-19
15. Related Business Rules (Appendix-C)	BR-EXM-002
16. Related Security Controls	N/A

17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-SCAL-007 — Maximum Fee Transactions

1. NFR ID	NFR-SCAL-007
2. Title	Maximum Fee Transactions
3. Category	Scalability
4. Description	Maximum fee transaction records the system must store and query per academic year.
5. Business Objective	Support high-volume collection periods (e.g., start-of-term payment surges).
6. Rationale	Financial reporting (FR-26) performance is sensitive to transaction table size.
7. Acceptance Criteria	Financial reports generate within Performance NFR targets at this volume.
8. Measurement Method	Load testing with synthetic data
9. Target Value	500,000+ transactions per academic year
10. Priority	High
11. Risk	Medium
12. Dependencies	Database Performance
13. Applicable Modules	Fee Collection
14. Related FRs (Appendix-E)	FR-23, FR-24, FR-26
15. Related Business Rules (Appendix-C)	BR-FEE-001
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard

21. Remarks	No outstanding client input required for this NFR.
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NFR-SCAL-008 — Database & Storage Growth

1. NFR ID	NFR-SCAL-008
2. Title	Database & Storage Growth
3. Category	Scalability
4. Description	Overall architecture capacity to accommodate multi-year data growth without redesign.
5. Business Objective	Avoid a costly re-architecture as historical data accumulates (RGD v2.0 Section 34, Data Retention).
6. Rationale	Combined with the Data Retention NFR, total storage grows predictably and must be planned for.
7. Acceptance Criteria	Architecture review confirms storage scaling path (e.g., horizontal partitioning, archival tiering) is defined before go-live.
8. Measurement Method	Architecture/capacity planning review
9. Target Value	5-year data growth horizon without redesign
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Data Retention (NFR-RET-001), Archiving (NFR-ARC-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic load/stress testing at 2x target capacity
18. Validation Method	Capacity planning review
19. Verification Method	Load/stress testing at 2x target capacity
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Availability (2 NFRs)**NFR-AVAIL-001 — System Uptime**

1. NFR ID	NFR-AVAIL-001
2. Title	System Uptime
3. Category	Availability
4. Description	Overall system availability during academic operating hours.
5. Business Objective	Ensure the ERP is available when the school actually needs it.
6. Rationale	Consistent with RGD v2.0 Section 19 Non-Functional Requirements (Availability).
7. Acceptance Criteria	Uptime monitoring confirms target met over each rolling 90-day period.
8. Measurement Method	Uptime monitoring (rolling 90-day window)
9. Target Value	99.5% during academic operating hours
10. Priority	Critical
11. Risk	High
12. Dependencies	Disaster Recovery, Monitoring series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	SLA report review
19. Verification Method	Uptime monitoring over a rolling 90-day window
20. Compliance Standard	Internal SLA
21. Remarks	No outstanding client input required for this NFR.

NFR-AVAIL-002 — Planned Maintenance Window

1. NFR ID	NFR-AVAIL-002
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2. Title	Planned Maintenance Window
3. Category	Availability
4. Description	Scheduled outage allowance for maintenance/upgrades.
5. Business Objective	Balance system evolution needs against operational disruption.
6. Rationale	Unannounced outages during school hours are highly disruptive to Attendance and Fee Collection.
7. Acceptance Criteria	Maintenance windows are scheduled off-peak with advance notice; verified via maintenance log review.
8. Measurement Method	Maintenance log audit
9. Target Value	≤4 hours/month, ≥72-hour advance notice
10. Priority	Medium
11. Risk	Low
12. Dependencies	Business Continuity (NFR-BC-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	SLA report review
19. Verification Method	Uptime monitoring over a rolling 90-day window
20. Compliance Standard	Internal SLA
21. Remarks	No outstanding client input required for this NFR.

Category: Reliability (2 NFRs)**NFR-REL-001 — Transaction Error Rate**

1. NFR ID	NFR-REL-001
2. Title	Transaction Error Rate
3. Category	Reliability
4. Description	Failure rate for critical transactions (fee payment, attendance save, exam lock).
5. Business Objective	Ensure the system is trustworthy for financially and academically critical actions.
6. Rationale	A high error rate on fee payment directly threatens revenue integrity.
7. Acceptance Criteria	Error rate measured and reported monthly against target.
8. Measurement Method	Application error-rate monitoring
9. Target Value	<0.1% for critical transactions
10. Priority	Critical
11. Risk	High
12. Dependencies	Monitoring series, Data Integrity series
13. Applicable Modules	Fee Collection, Attendance, Examination
14. Related FRs (Appendix-E)	FR-10, FR-23, FR-24, FR-18
15. Related Business Rules (Appendix-C)	BR-FEE-001, BR-ATT-001, BR-EXM-002
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic fault-injection / chaos testing
18. Validation Method	Failure-scenario test suite
19. Verification Method	Fault-injection / chaos testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-REL-002 — Recovery from Transient Failure

1. NFR ID	NFR-REL-002
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2. Title	Recovery from Transient Failure
3. Category	Reliability
4. Description	Automatic recovery/retry behavior for transient failures (network blips, brief service interruption).
5. Business Objective	Prevent data loss or duplicate transactions during minor, recoverable failures.
6. Rationale	Directly supports resolution of the BR-ADM-001/BR-ADM-007 race-condition risk and BR-FEE-006 duplicate-payment prevention.
7. Acceptance Criteria	Fault-injection testing confirms no data loss or duplication across simulated transient failures.
8. Measurement Method	Fault-injection / chaos testing
9. Target Value	Zero data loss or duplicate transactions on transient failure
10. Priority	High
11. Risk	High
12. Dependencies	Data Integrity (NFR-DI-002)
13. Applicable Modules	Admission & Enrollment, Fee Collection
14. Related FRs (Appendix-E)	FR-02, FR-24
15. Related Business Rules (Appendix-C)	BR-ADM-001, BR-FEE-006
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic fault-injection / chaos testing
18. Validation Method	Failure-scenario test suite
19. Verification Method	Fault-injection / chaos testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Security (3 NFRs)**NFR-SEC-001 — CSRF Protection**

1. NFR ID	NFR-SEC-001
2. Title	CSRF Protection
3. Category	Security
4. Description	Protection against Cross-Site Request Forgery on all state-changing requests.
5. Business Objective	Prevent unauthorized actions performed on behalf of an authenticated user.
6. Rationale	A core OWASP Top 10 risk category; especially relevant given Fee Collection and Examination modules perform sensitive state changes.
7. Acceptance Criteria	Penetration test confirms no exploitable CSRF vector on any state-changing endpoint.
8. Measurement Method	Penetration testing
9. Target Value	Zero exploitable CSRF vulnerabilities
10. Priority	Critical
11. Risk	High
12. Dependencies	Session Management (NFR-SESS-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Anti-CSRF tokens on all state-changing requests
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic penetration testing and automated security scanning (sast/dast)
18. Validation Method	Security code review
19. Verification Method	Penetration testing and automated security scanning (SAST/DAST)
20. Compliance Standard	OWASP Top 10
21. Remarks	No outstanding client input required for this NFR.

NFR-SEC-002 — XSS Protection

1. NFR ID	NFR-SEC-002
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2. Title	XSS Protection
3. Category	Security
4. Description	Protection against Cross-Site Scripting via input sanitization/output encoding.
5. Business Objective	Prevent malicious script injection through user-supplied input (e.g., messaging, circulars).
6. Rationale	Communication module (FR-37, FR-38, FR-39) is a plausible XSS vector given free-text message content.
7. Acceptance Criteria	Penetration test and automated scan confirm no exploitable XSS vector.
8. Measurement Method	SAST/DAST automated scanning + penetration testing
9. Target Value	Zero exploitable XSS vulnerabilities
10. Priority	Critical
11. Risk	High
12. Dependencies	File Upload Security (NFR-FUS-001)
13. Applicable Modules	Communication & Notifications, All Modules
14. Related FRs (Appendix-E)	FR-37, FR-38, FR-39
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Input sanitization, output encoding, Content Security Policy
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic penetration testing and automated security scanning (sast/dast)
18. Validation Method	Security code review
19. Verification Method	Penetration testing and automated security scanning (SAST/DAST)
20. Compliance Standard	OWASP Top 10
21. Remarks	No outstanding client input required for this NFR.

NFR-SEC-003 — SQL Injection Protection

1. NFR ID	NFR-SEC-003
2. Title	SQL Injection Protection
3. Category	Security
4. Description	Protection against SQL Injection via parameterized queries/ORM usage.

5. Business Objective	Protect the integrity and confidentiality of all system data.
6. Rationale	A single SQL injection vulnerability could expose the entire student/financial database.
7. Acceptance Criteria	Automated scan and penetration test confirm no exploitable SQL injection vector.
8. Measurement Method	SAST/DAST automated scanning + penetration testing
9. Target Value	Zero exploitable SQL injection vulnerabilities
10. Priority	Critical
11. Risk	High
12. Dependencies	Data Integrity series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Parameterized queries / ORM, no dynamic SQL concatenation
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic penetration testing and automated security scanning (sast/dast)
18. Validation Method	Security code review
19. Verification Method	Penetration testing and automated security scanning (SAST/DAST)
20. Compliance Standard	OWASP Top 10
21. Remarks	No outstanding client input required for this NFR.

Category: Authentication (2 NFRs)**NFR-AUTHN-001 — Login Mechanism Security**

1. NFR ID	NFR-AUTHN-001
2. Title	Login Mechanism Security
3. Category	Authentication
4. Description	Secure username/password authentication with account lockout protection.
5. Business Objective	Prevent credential-stuffing and brute-force login attacks.
6. Rationale	Front door of the system; weak authentication undermines every other security control.
7. Acceptance Criteria	Penetration test confirms lockout triggers correctly and no bypass exists.
8. Measurement Method	Penetration testing
9. Target Value	Lockout after 5 failed attempts (see NFR-PWD-003)
10. Priority	Critical
11. Risk	High
12. Dependencies	Password Policy series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Account lockout, rate-limited login endpoint
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic penetration testing of login/session flows
18. Validation Method	Security code review
19. Verification Method	Penetration testing of login/session flows
20. Compliance Standard	OWASP ASVS
21. Remarks	No outstanding client input required for this NFR.

NFR-AUTHN-002 — Multi-Factor Authentication Readiness

1. NFR ID	NFR-AUTHN-002
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2. Title	Multi-Factor Authentication Readiness
3. Category	Authentication
4. Description	System architecture supports adding MFA (OTP/authenticator app) without redesign, even if not enabled at go-live.
5. Business Objective	Provide a compliance and security upgrade path as the institution's risk posture evolves.
6. Rationale	Financial (Fee Collection) and HR/Payroll modules justify a future MFA requirement for elevated roles.
7. Acceptance Criteria	Architecture review confirms an MFA hook point exists in the authentication flow.
8. Measurement Method	Architecture review
9. Target Value	<i>MFA-ready architecture; MFA itself is a future enhancement (Client/Product Decision Required on activation timeline)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Authentication (NFR-AUTHN-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	MFA-capable authentication framework
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic penetration testing of login/session flows
18. Validation Method	Security code review
19. Verification Method	Penetration testing of login/session flows
20. Compliance Standard	OWASP ASVS
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Authorization (1 NFR)**NFR-AUTHZ-001 — Role-Based Access Control Enforcement**

1. NFR ID	NFR-AUTHZ-001
2. Title	Role-Based Access Control Enforcement
3. Category	Authorization
4. Description	Every module enforces RBAC consistent with the Role Permissions defined per requirement in Appendix-E.
5. Business Objective	Ensure only authorized roles can perform Create/Read/Update/Delete/Approve/Reject/Export/Print actions.
6. Rationale	Directly operationalizes Appendix-E Field 38 and BR-RPT-001-style authorization patterns across all modules.
7. Acceptance Criteria	Access-control test matrix confirms every role boundary defined in Appendix-E Field 38 is enforced.
8. Measurement Method	Role-based access test matrix execution
9. Target Value	100% of Appendix-E Field 38 permission boundaries enforced
10. Priority	Critical
11. Risk	High
12. Dependencies	Authentication series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	BR-FEE-002, BR-RPT-001, BR-COM-002
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic role-based access test matrix execution
18. Validation Method	Cross-reference against Appendix-E Field 38 Role Permissions
19. Verification Method	Role-based access test matrix execution
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Audit Logging (1 NFR)**NFR-AUDIT-001 — Audit Trail Completeness**

1. NFR ID	NFR-AUDIT-001
2. Title	Audit Trail Completeness
3. Category	Audit Logging
4. Description	Every create/update/delete/approve/override action is logged per the standard audit template.
5. Business Objective	Support forensic investigation, dispute resolution, and regulatory audit requests.
6. Rationale	Directly operationalizes Appendix-E Field 24 (Audit Log Requirements) across all 47 functional requirements.
7. Acceptance Criteria	Sampling audit confirms 100% of tested actions produce a complete audit record (Who/When/Old Value/New Value/IP/Reason).
8. Measurement Method	Audit-log completeness sampling audit
9. Target Value	100% of state-changing actions logged
10. Priority	Critical
11. Risk	High
12. Dependencies	Logging series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	All immutability-related Business Rules (e.g., BR-SIS-004, BR-EXM-003, BR-FEE-001)
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24
19. Verification Method	Audit-log completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

Category: Data Integrity (2 NFRs)**NFR-DI-001 — Referential Integrity Enforcement**

1. NFR ID	NFR-DI-001
2. Title	Referential Integrity Enforcement
3. Category	Data Integrity
4. Description	Database enforces foreign-key constraints preventing orphaned records across modules.
5. Business Objective	Prevent data corruption such as a Fee ledger entry referencing a non-existent student.
6. Rationale	Directly supports the cross-module data lineage documented in Appendix-D's Cross-Module Dependency Matrix.
7. Acceptance Criteria	Constraint test suite confirms orphaned-record creation is impossible at the database layer.
8. Measurement Method	Database constraint test suite
9. Target Value	Zero orphaned records possible
10. Priority	Critical
11. Risk	High
12. Dependencies	Database Performance
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic database constraint and transaction test suite
18. Validation Method	Schema/constraint review
19. Verification Method	Database constraint and transaction test suite
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-DI-002 — Transactional Consistency

1. NFR ID	NFR-DI-002
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2. Title	Transactional Consistency
3. Category	Data Integrity
4. Description	Multi-step operations execute atomically or fully roll back, with no partial state possible.
5. Business Objective	Prevent the race-condition class of risk identified in Appendix-C Section 3.2 (BR-ADM-001/BR-ADM-007).
6. Rationale	A partial transaction (e.g., seat decremented but admission not confirmed) corrupts downstream capacity tracking.
7. Acceptance Criteria	Fault-injection testing confirms no partial-state outcome across simulated mid-transaction failures.
8. Measurement Method	Fault-injection / transaction testing
9. Target Value	Zero partial-state transactions
10. Priority	Critical
11. Risk	High
12. Dependencies	Reliability (NFR-REL-002)
13. Applicable Modules	Admission & Enrollment, Fee Collection, HR & Payroll
14. Related FRs (Appendix-E)	FR-02, FR-04, FR-24, FR-34
15. Related Business Rules (Appendix-C)	BR-ADM-001, BR-ADM-007, BR-HR-003
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic database constraint and transaction test suite
18. Validation Method	Schema/constraint review
19. Verification Method	Database constraint and transaction test suite
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Backup (5 NFRs)**NFR-BKP-001 — Backup Frequency**

1. NFR ID	NFR-BKP-001
2. Title	Backup Frequency
3. Category	Backup
4. Description	Automated backup schedule for full database and transaction logs.
5. Business Objective	Minimize potential data loss window in the event of failure.
6. Rationale	Directly determines the achievable Recovery Point Objective (NFR-BKP-004).
7. Acceptance Criteria	Backup logs confirm scheduled backups complete successfully on schedule.
8. Measurement Method	Backup log review
9. Target Value	Daily full backup + continuous transaction log backup
10. Priority	Critical
11. Risk	High
12. Dependencies	Disaster Recovery series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic scheduled restore-test execution
18. Validation Method	Backup log review
19. Verification Method	Scheduled restore-test execution
20. Compliance Standard	ISO 27001 Annex A.12.3
21. Remarks	No outstanding client input required for this NFR.

NFR-BKP-002 — Backup Retention Period

1. NFR ID	NFR-BKP-002
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2. Title	Backup Retention Period
3. Category	Backup
4. Description	Duration for which backups are retained before deletion.
5. Business Objective	Balance recovery flexibility against storage cost and Data Retention policy alignment.
6. Rationale	Retention period must be confirmed alongside the Data Retention NFR (NFR-RET-001) for policy consistency.
7. Acceptance Criteria	Backup retention configuration matches the agreed policy.
8. Measurement Method	Backup configuration audit
9. Target Value	<i>30 days rolling + monthly archive for 1 year (Client/Product Decision Required to confirm exact policy)</i>
10. Priority	High
11. Risk	Medium
12. Dependencies	Data Retention (NFR-RET-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic scheduled restore-test execution
18. Validation Method	Backup log review
19. Verification Method	Scheduled restore-test execution
20. Compliance Standard	ISO 27001 Annex A.12.3
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-BKP-003 — Recovery Time Objective (RTO)

1. NFR ID	NFR-BKP-003
2. Title	Recovery Time Objective (RTO)
3. Category	Backup
4. Description	Maximum acceptable time to fully restore the system after a failure.
5. Business Objective	Bound the operational disruption window for the school in a worst-case scenario.

6. Rationale	Directly informs the Disaster Recovery process design (NFR-DR-001).
7. Acceptance Criteria	DR drill confirms restoration completes within target.
8. Measurement Method	DR drill timing measurement
9. Target Value	<i>≤4 hours (Client/Product Decision Required to confirm against client's operational tolerance)</i>
10. Priority	Critical
11. Risk	High
12. Dependencies	Disaster Recovery (NFR-DR-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic scheduled restore-test execution
18. Validation Method	Backup log review
19. Verification Method	Scheduled restore-test execution
20. Compliance Standard	ISO 27001 Annex A.12.3
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-BKP-004 — Recovery Point Objective (RPO)

1. NFR ID	NFR-BKP-004
2. Title	Recovery Point Objective (RPO)
3. Category	Backup
4. Description	Maximum acceptable data loss, measured in time, in the event of a failure.
5. Business Objective	Bound the worst-case data-loss exposure (e.g., attendance/fee transactions since the last backup).
6. Rationale	Directly informs backup frequency (NFR-BKP-001) design.
7. Acceptance Criteria	DR drill confirms data loss does not exceed target in a simulated failure.
8. Measurement Method	DR drill data-loss measurement
9. Target Value	<i>≤1 hour (Client/Product Decision Required to confirm against client's risk tolerance)</i>

10. Priority	Critical
11. Risk	High
12. Dependencies	Backup Frequency (NFR-BKP-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic scheduled restore-test execution
18. Validation Method	Backup log review
19. Verification Method	Scheduled restore-test execution
20. Compliance Standard	ISO 27001 Annex A.12.3
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-BKP-005 — Backup Integrity Verification

1. NFR ID	NFR-BKP-005
2. Title	Backup Integrity Verification
3. Category	Backup
4. Description	Periodic test restoration to confirm backups are actually usable, not merely present.
5. Business Objective	A backup that cannot be restored provides false assurance.
6. Rationale	This is the single most commonly overlooked backup requirement in enterprise systems.
7. Acceptance Criteria	Monthly restore test completes successfully and is logged.
8. Measurement Method	Scheduled restore-test execution
9. Target Value	Monthly automated restore test
10. Priority	High
11. Risk	Medium
12. Dependencies	Backup Frequency (NFR-BKP-001)
13. Applicable Modules	All Modules

14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic scheduled restore-test execution
18. Validation Method	Backup log review
19. Verification Method	Scheduled restore-test execution
20. Compliance Standard	ISO 27001 Annex A.12.3
21. Remarks	No outstanding client input required for this NFR.

Category: Disaster Recovery (2 NFRs)**NFR-DR-001 — Disaster Recovery Process**

1. NFR ID	NFR-DR-001
2. Title	Disaster Recovery Process
3. Category	Disaster Recovery
4. Description	Documented, tested failover procedure to a secondary region/data center.
5. Business Objective	Ensure business-critical operations can resume after a major infrastructure failure.
6. Rationale	Without a tested (not just documented) process, RTO/RPO targets are aspirational rather than achievable.
7. Acceptance Criteria	DR runbook exists, is version-controlled, and has been executed at least once successfully.
8. Measurement Method	DR drill execution and runbook review
9. Target Value	Documented and drill-tested failover procedure
10. Priority	Critical
11. Risk	High
12. Dependencies	Backup series, Business Continuity (NFR-BC-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	DR runbook review
19. Verification Method	Full DR drill execution
20. Compliance Standard	ISO 22301
21. Remarks	No outstanding client input required for this NFR.

NFR-DR-002 — DR Drill Frequency

1. NFR ID	NFR-DR-002
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2. Title	DR Drill Frequency
3. Category	Disaster Recovery
4. Description	Cadence at which a full disaster recovery drill is conducted.
5. Business Objective	Ensure the DR process remains valid as the system evolves, not just at initial design.
6. Rationale	An untested DR process degrades in reliability as infrastructure and data volume change over time.
7. Acceptance Criteria	DR drill log confirms at least one full drill per year with results documented.
8. Measurement Method	DR drill log audit
9. Target Value	At least annually
10. Priority	High
11. Risk	Medium
12. Dependencies	DR Process (NFR-DR-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	DR runbook review
19. Verification Method	Full DR drill execution
20. Compliance Standard	ISO 22301
21. Remarks	No outstanding client input required for this NFR.

Category: Business Continuity (1 NFR)**NFR-BC-001 — Continuity of Critical Operations**

1. NFR ID	NFR-BC-001
2. Title	Continuity of Critical Operations
3. Category	Business Continuity
4. Description	Manual-workaround procedures are documented for Admission, Attendance, and Fee Collection during extended outages.
5. Business Objective	Ensure the school can continue essential operations even during a prolonged system outage.
6. Rationale	Directly reuses the Workaround field (Field 14) already documented for each requirement in Appendix-D.
7. Acceptance Criteria	Workaround procedures exist and are accessible to Admin Staff/Finance Team independent of system availability (e.g., printed procedure).
8. Measurement Method	Business continuity tabletop exercise
9. Target Value	Documented workaround for all Critical-priority requirements
10. Priority	High
11. Risk	Medium
12. Dependencies	Availability series, Disaster Recovery series
13. Applicable Modules	Admission & Enrollment, Attendance Management, Fee Collection
14. Related FRs (Appendix-E)	FR-01, FR-10, FR-23
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Manual-workaround procedure review (Appendix-A)
19. Verification Method	Business continuity tabletop exercise
20. Compliance Standard	ISO 22301
21. Remarks	No outstanding client input required for this NFR.

Category: Usability (2 NFRs)

NFR-USA-001 — Task Completion Efficiency

1. NFR ID	NFR-USA-001
2. Title	Task Completion Efficiency
3. Category	Usability
4. Description	A trained Admin Staff/Teacher user can complete common tasks within a defined number of steps.
5. Business Objective	Minimize daily operational friction for high-frequency tasks (e.g., marking attendance for a class).
6. Rationale	Directly supports RGD v2.0 OBJ around reducing administrative overhead.
7. Acceptance Criteria	Usability test confirms representative users complete common tasks (e.g., mark attendance for a class) within the target step count.
8. Measurement Method	Usability testing with representative users
9. Target Value	<i>Attendance marking for a class in ≤5 clicks/steps (Client/Product Decision Required to confirm exact target)</i>
10. Priority	Medium
11. Risk	Low
12. Dependencies	N/A
13. Applicable Modules	Attendance Management, Fee Collection
14. Related FRs (Appendix-E)	FR-10, FR-23
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic usability testing with representative admin staff/teacher users
18. Validation Method	Task-completion time study
19. Verification Method	Usability testing with representative Admin Staff/Teacher users
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-USA-002 — New User Training Time

1. NFR ID	NFR-USA-002
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2. Title	New User Training Time
3. Category	Usability
4. Description	Time for a new staff user to reach basic operational proficiency.
5. Business Objective	Minimize onboarding cost and resistance to adoption.
6. Rationale	Directly relevant to the Communication and change-management aspects of go-live.
7. Acceptance Criteria	Training study confirms representative new users reach basic proficiency within target.
8. Measurement Method	Training-time study with new users
9. Target Value	<i>≤2 hours guided training (Client/Product Decision Required to confirm target)</i>
10. Priority	Medium
11. Risk	Low
12. Dependencies	N/A
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic usability testing with representative admin staff/teacher users
18. Validation Method	Task-completion time study
19. Verification Method	Usability testing with representative Admin Staff/Teacher users
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Accessibility (5 NFRs)**NFR-ACC-001 — WCAG 2.1 AA Conformance**

1. NFR ID	NFR-ACC-001
2. Title	WCAG 2.1 AA Conformance
3. Category	Accessibility
4. Description	Public-facing portals (parent/student) conform to WCAG 2.1 Level AA.
5. Business Objective	Ensure equitable access for users with disabilities; increasingly a legal/procurement expectation.
6. Rationale	No accessibility requirement existed in RGD v2.0; this closes that gap ahead of Architecture Design.
7. Acceptance Criteria	Automated and manual accessibility audit confirms Level AA conformance across all public-facing pages.
8. Measurement Method	Automated scan (axe/WAVE) + manual audit
9. Target Value	WCAG 2.1 Level AA
10. Priority	High
11. Risk	Medium
12. Dependencies	Keyboard Navigation, Screen Reader, Color Contrast (below)
13. Applicable Modules	Parent/Student Portal (all modules with portal exposure)
14. Related FRs (Appendix-E)	FR-21, FR-29, FR-38, FR-39, FR-40
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated accessibility scan (axe/wave) plus manual screen-reader testing
18. Validation Method	WCAG 2.1 AA checklist audit
19. Verification Method	Automated accessibility scan (axe/WAVE) plus manual screen-reader testing
20. Compliance Standard	WCAG 2.1 Level AA
21. Remarks	No outstanding client input required for this NFR.

NFR-ACC-002 — Keyboard Navigation

1. NFR ID	NFR-ACC-002
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2. Title	Keyboard Navigation
3. Category	Accessibility
4. Description	All interactive elements are operable via keyboard alone, without requiring a mouse.
5. Business Objective	Support users who cannot use a pointing device, and screen-reader users generally.
6. Rationale	A core, foundational WCAG 2.1 AA success criterion.
7. Acceptance Criteria	Manual keyboard-only walkthrough confirms all interactive elements are reachable and operable.
8. Measurement Method	Manual keyboard-only testing
9. Target Value	100% of interactive elements keyboard-operable
10. Priority	High
11. Risk	Low
12. Dependencies	WCAG 2.1 AA (NFR-ACC-001)
13. Applicable Modules	Parent/Student Portal
14. Related FRs (Appendix-E)	FR-21, FR-29, FR-38, FR-39
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated accessibility scan (axe/wave) plus manual screen-reader testing
18. Validation Method	WCAG 2.1 AA checklist audit
19. Verification Method	Automated accessibility scan (axe/WAVE) plus manual screen-reader testing
20. Compliance Standard	WCAG 2.1 Level AA
21. Remarks	No outstanding client input required for this NFR.

NFR-ACC-003 — Screen Reader Support

1. NFR ID	NFR-ACC-003
2. Title	Screen Reader Support
3. Category	Accessibility
4. Description	All content is compatible with common screen readers (NVDA, JAWS, VoiceOver).

5. Business Objective	Support visually impaired parents/students/staff.
6. Rationale	Directly required for WCAG 2.1 AA conformance (NFR-ACC-001).
7. Acceptance Criteria	Manual screen-reader walkthrough confirms all content is correctly announced and navigable.
8. Measurement Method	Manual screen-reader testing
9. Target Value	Full compatibility with NVDA, JAWS, and VoiceOver
10. Priority	Medium
11. Risk	Low
12. Dependencies	WCAG 2.1 AA (NFR-ACC-001)
13. Applicable Modules	Parent/Student Portal
14. Related FRs (Appendix-E)	FR-21, FR-29, FR-38, FR-39
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated accessibility scan (axe/wave) plus manual screen-reader testing
18. Validation Method	WCAG 2.1 AA checklist audit
19. Verification Method	Automated accessibility scan (axe/WAVE) plus manual screen-reader testing
20. Compliance Standard	WCAG 2.1 Level AA
21. Remarks	No outstanding client input required for this NFR.

NFR-ACC-004 — Color Contrast

1. NFR ID	NFR-ACC-004
2. Title	Color Contrast
3. Category	Accessibility
4. Description	Minimum color contrast ratio for text against its background.
5. Business Objective	Ensure readability for users with low vision or color-vision deficiency.
6. Rationale	A specific, objectively testable WCAG 2.1 AA success criterion (1.4.3).
7. Acceptance Criteria	Automated contrast-checker confirms all text meets the minimum ratio.

8. Measurement Method	Automated contrast scanning
9. Target Value	≥4.5:1 for normal text (WCAG 2.1 AA)
10. Priority	Medium
11. Risk	Low
12. Dependencies	WCAG 2.1 AA (NFR-ACC-001)
13. Applicable Modules	All Modules (UI-wide)
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated accessibility scan (axe/wave) plus manual screen-reader testing
18. Validation Method	WCAG 2.1 AA checklist audit
19. Verification Method	Automated accessibility scan (axe/WAVE) plus manual screen-reader testing
20. Compliance Standard	WCAG 2.1 Level AA
21. Remarks	No outstanding client input required for this NFR.

NFR-ACC-005 — Responsive Design

1. NFR ID	NFR-ACC-005
2. Title	Responsive Design
3. Category	Accessibility
4. Description	UI adapts correctly across desktop, tablet, and mobile viewport sizes.
5. Business Objective	Support parents/staff accessing the system from varied devices (RGD v2.0 NFR Compatibility).
6. Rationale	Explicitly required in RGD v2.0 Section 19 (responsive design for desktop, tablet, and mobile).
7. Acceptance Criteria	Manual QA confirms correct layout/functionality across a defined device/viewport matrix.
8. Measurement Method	Manual responsive-design QA
9. Target Value	Correct rendering at desktop, tablet, and mobile breakpoints
10. Priority	High

11. Risk	Medium
12. Dependencies	Mobile Compatibility (NFR-MOB-001), Browser Compatibility (NFR-BRC-001)
13. Applicable Modules	All Modules (UI-wide)
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated accessibility scan (axe/wave) plus manual screen-reader testing
18. Validation Method	WCAG 2.1 AA checklist audit
19. Verification Method	Automated accessibility scan (axe/WAVE) plus manual screen-reader testing
20. Compliance Standard	WCAG 2.1 Level AA
21. Remarks	No outstanding client input required for this NFR.

Category: Browser Compatibility (1 NFR)**NFR-BRC-001 — Supported Browser Matrix**

1. NFR ID	NFR-BRC-001
2. Title	Supported Browser Matrix
3. Category	Browser Compatibility
4. Description	System functions correctly on the current and immediately prior major version of Chrome, Edge, Firefox, and Safari.
5. Business Objective	Avoid excluding users on common, actively maintained browsers.
6. Rationale	Consistent with RGD v2.0 Section 19 device/browser compatibility requirement.
7. Acceptance Criteria	Cross-browser QA confirms full functional parity across the supported matrix.
8. Measurement Method	Cross-browser test matrix execution
9. Target Value	Chrome, Edge, Firefox, Safari — current + prior major version
10. Priority	Medium
11. Risk	Low
12. Dependencies	Responsive Design (NFR-ACC-005)
13. Applicable Modules	All Modules (UI-wide)
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic cross-browser test matrix execution
18. Validation Method	Manual QA across supported browser list
19. Verification Method	Cross-browser test matrix execution
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Mobile Compatibility (1 NFR)**NFR-MOB-001 — Mobile Browser Functional Parity**

1. NFR ID	NFR-MOB-001
2. Title	Mobile Browser Functional Parity
3. Category	Mobile Compatibility
4. Description	Full functional parity on Android and iOS mobile browsers.
5. Business Objective	Support parents, who are the highest-volume mobile user group, accessing the portal on the go.
6. Rationale	Explicitly required per RGD v2.0 Section 19.
7. Acceptance Criteria	Manual QA on representative Android/iOS devices confirms parity with desktop functionality.
8. Measurement Method	Cross-device mobile test matrix execution
9. Target Value	Full functional parity, Android and iOS
10. Priority	High
11. Risk	Medium
12. Dependencies	Responsive Design (NFR-ACC-005)
13. Applicable Modules	Parent/Student Portal
14. Related FRs (Appendix-E)	FR-05, FR-11, FR-21, FR-24, FR-31, FR-38, FR-39
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic cross-device mobile test matrix execution
18. Validation Method	Manual QA on representative Android/iOS devices
19. Verification Method	Cross-device mobile test matrix execution
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: API Performance (1 NFR)

NFR-APIP-001 — API Throughput & Rate Limiting

1. NFR ID	NFR-APIP-001
2. Title	API Throughput & Rate Limiting
3. Category	API Performance
4. Description	Internal APIs support a defined sustained throughput with rate-limiting to prevent abuse.
5. Business Objective	Protect system stability from both legitimate peak load and abusive/erroneous client behavior.
6. Rationale	Distinct from API response time (NFR-PERF-011): this governs sustained volume, not single-call latency.
7. Acceptance Criteria	Load test confirms target throughput is sustained without error-rate degradation; rate-limit correctly rejects excess calls.
8. Measurement Method	API load testing with rate-limit boundary testing
9. Target Value	<i>Sustained throughput target Client/Product Decision Required; rate-limiting enabled on all public endpoints</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	API Response Time (NFR-PERF-011), Network Security (NFR-NET-002)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	API rate limiting, request throttling
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic api load testing with rate-limit boundary testing
18. Validation Method	API test suite execution
19. Verification Method	API load testing with rate-limit boundary testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Database Performance (1 NFR)**NFR-DBP-001 — Query Optimization & Indexing**

1. NFR ID	NFR-DBP-001
2. Title	Query Optimization & Indexing
3. Category	Database Performance
4. Description	All high-frequency queries (attendance, fee, exam) are covered by appropriate database indexes.
5. Business Objective	Prevent performance decay as data volume grows across academic years (see Scalability series).
6. Rationale	Underpins nearly every Performance NFR — indexing is the primary lever for query-response targets.
7. Acceptance Criteria	Query-plan review confirms no full-table scans on high-frequency query paths at target data volume.
8. Measurement Method	Query-plan review under load
9. Target Value	100% of high-frequency queries index-covered
10. Priority	High
11. Risk	High
12. Dependencies	Scalability series, Database Query Response (NFR-PERF-012)
13. Applicable Modules	Attendance Management, Fee Collection, Examination
14. Related FRs (Appendix-E)	FR-10, FR-13, FR-23, FR-26, FR-18, FR-19
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Index coverage audit
19. Verification Method	Query execution-plan review under production-scale data volume
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Caching (1 NFR)

NFR-CACHE-001 — Dashboard & Master Data Caching

1. NFR ID	NFR-CACHE-001
2. Title	Dashboard & Master Data Caching
3. Category	Caching
4. Description	Frequently accessed, slow-changing data (master data, published timetables, dashboards) is cached with defined invalidation triggers.
5. Business Objective	Reduce database load and improve perceived responsiveness for high-traffic read paths.
6. Rationale	Directly supports Dashboard Loading (NFR-PERF-003) and Page Load Time (NFR-PERF-001) targets.
7. Acceptance Criteria	Cache hit-rate monitoring confirms target hit rate under load; invalidation test confirms stale data is never served beyond the defined window.
8. Measurement Method	Cache hit-rate monitoring + invalidation test suite
9. Target Value	<i>≥80% cache hit rate for cacheable read paths (Client/Product Decision Required to confirm target)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Dashboard Loading (NFR-PERF-003), Page Load Time (NFR-PERF-001)
13. Applicable Modules	Reports & Analytics, Timetable & Scheduling
14. Related FRs (Appendix-E)	FR-14, FR-40
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic cache hit-rate monitoring under load testing
18. Validation Method	Cache invalidation test suite
19. Verification Method	Cache hit-rate monitoring under load testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Session Management (1 NFR)

NFR-SESS-001 — Session Timeout

1. NFR ID	NFR-SESS-001
2. Title	Session Timeout
3. Category	Session Management
4. Description	Inactive user sessions automatically expire after a configurable idle period.
5. Business Objective	Reduce the window of exposure from an unattended, logged-in session (e.g., a shared front-office computer).
6. Rationale	A standard OWASP ASVS session-management control, especially relevant for shared-device use in Admin offices.
7. Acceptance Criteria	Security test confirms a session is invalidated after the configured idle period and requires re-authentication.
8. Measurement Method	Session-timeout test cases
9. Target Value	<i>30 minutes idle timeout (Client/Product Decision Required to confirm policy, may vary by role)</i>
10. Priority	High
11. Risk	Medium
12. Dependencies	Authentication (NFR-AUTHN-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Server-side session expiry, secure session token handling
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic session-timeout and token-expiry test cases
18. Validation Method	Security code review
19. Verification Method	Session-timeout and token-expiry test cases
20. Compliance Standard	OWASP ASVS
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Password Policy (3 NFRs)**NFR-PWD-001 — Password Complexity**

1. NFR ID	NFR-PWD-001
2. Title	Password Complexity
3. Category	Password Policy
4. Description	Minimum password complexity requirements at account creation and change.
5. Business Objective	Reduce the risk of trivially guessable passwords.
6. Rationale	A baseline security control expected in any enterprise system handling financial and student PII.
7. Acceptance Criteria	Password-policy enforcement test confirms non-compliant passwords are rejected at creation/change.
8. Measurement Method	Automated password-policy enforcement test
9. Target Value	Minimum 8 characters, mixed case, number, special character
10. Priority	High
11. Risk	Medium
12. Dependencies	Authentication (NFR-AUTHN-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Password complexity validation
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated password-policy enforcement test cases
18. Validation Method	Security code review
19. Verification Method	Automated password-policy enforcement test cases
20. Compliance Standard	OWASP ASVS
21. Remarks	No outstanding client input required for this NFR.

NFR-PWD-002 — Password Expiry

1. NFR ID	NFR-PWD-002
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2. Title	Password Expiry
3. Category	Password Policy
4. Description	Mandatory password change after a configured interval.
5. Business Objective	Limit the useful lifetime of a compromised-but-undetected credential.
6. Rationale	Policy value should be balanced against usability; overly frequent expiry can encourage weak password reuse patterns.
7. Acceptance Criteria	Test confirms users are prompted to change password upon reaching the configured interval.
8. Measurement Method	Automated password-policy enforcement test
9. Target Value	<i>90 days (Client/Product Decision Required to confirm policy; note modern guidance increasingly favors longer intervals with MFA over frequent forced rotation)</i>
10. Priority	Medium
11. Risk	Low
12. Dependencies	MFA Readiness (NFR-AUTHN-002)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Password expiry enforcement
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated password-policy enforcement test cases
18. Validation Method	Security code review
19. Verification Method	Automated password-policy enforcement test cases
20. Compliance Standard	OWASP ASVS
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

NFR-PWD-003 — Account Lockout

1. NFR ID	NFR-PWD-003
2. Title	Account Lockout
3. Category	Password Policy
4. Description	Account is locked after a defined number of consecutive failed login attempts.

5. Business Objective	Mitigate brute-force credential-guessing attacks.
6. Rationale	Directly supports NFR-AUTHN-001 (Login Mechanism Security).
7. Acceptance Criteria	Test confirms lockout triggers at the configured threshold and the unlock procedure functions correctly.
8. Measurement Method	Automated lockout-trigger test
9. Target Value	Lockout after 5 consecutive failed attempts, with a defined unlock procedure
10. Priority	Critical
11. Risk	Medium
12. Dependencies	Login Mechanism Security (NFR-AUTHN-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Account lockout with defined unlock/reset procedure
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic automated password-policy enforcement test cases
18. Validation Method	Security code review
19. Verification Method	Automated password-policy enforcement test cases
20. Compliance Standard	OWASP ASVS
21. Remarks	No outstanding client input required for this NFR.

Category: Encryption (2 NFRs)**NFR-ENC-001 — Encryption at Rest**

1. NFR ID	NFR-ENC-001
2. Title	Encryption at Rest
3. Category	Encryption
4. Description	Sensitive data (PII, Aadhaar references, payment references, medical data) is encrypted at rest.
5. Business Objective	Protect data confidentiality in the event of unauthorized storage-layer access.
6. Rationale	Directly required for DPDP Act, 2023 compliance given the sensitivity of student/staff PII and medical data.
7. Acceptance Criteria	Encryption-configuration audit confirms all designated sensitive fields/tables are encrypted at rest.
8. Measurement Method	Encryption-configuration audit
9. Target Value	AES-256 (or equivalent) encryption at rest for all designated sensitive data
10. Priority	Critical
11. Risk	High
12. Dependencies	Data Privacy (NFR-PRIV-001)
13. Applicable Modules	Student Information Management, HR & Payroll, Fee Collection
14. Related FRs (Appendix-E)	FR-06, FR-33, FR-23
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Encryption at rest (AES-256 or equivalent)
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Security code review
19. Verification Method	Encryption-configuration audit (cipher suite, key management)
20. Compliance Standard	OWASP ASVS / DPDP Act, 2023
21. Remarks	No outstanding client input required for this NFR.

NFR-ENC-002 — Encryption in Transit

1. NFR ID	NFR-ENC-002
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2. Title	Encryption in Transit
3. Category	Encryption
4. Description	All data in transit between client and server, and between internal services, is encrypted.
5. Business Objective	Protect data confidentiality and integrity over the network.
6. Rationale	Overlaps with, and is enforced by, Network Security (NFR-NET-001); listed separately here per the requested category list.
7. Acceptance Criteria	TLS-configuration audit confirms all traffic is encrypted with no fallback to unencrypted transport.
8. Measurement Method	TLS/SSL configuration audit
9. Target Value	TLS 1.2 or higher for all traffic
10. Priority	Critical
11. Risk	High
12. Dependencies	TLS/HTTPS Enforcement (NFR-NET-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	TLS 1.2+ enforced network-wide
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Security code review
19. Verification Method	Encryption-configuration audit (cipher suite, key management)
20. Compliance Standard	OWASP ASVS / DPDP Act, 2023
21. Remarks	No outstanding client input required for this NFR.

Category: File Upload Security (1 NFR)**NFR-FUS-001 — File Upload Validation**

1. NFR ID	NFR-FUS-001
2. Title	File Upload Validation
3. Category	File Upload Security
4. Description	Uploaded files are validated for type, size, and scanned for malware before storage.
5. Business Objective	Prevent malicious file uploads via the Admission document-upload feature (FR-01) and similar upload points.
6. Rationale	FR-01 explicitly accepts applicant-supplied document uploads — an obvious attack surface if unvalidated.
7. Acceptance Criteria	Penetration test confirms malicious/oversized/wrong-type files are rejected before storage.
8. Measurement Method	Malicious-file-upload penetration testing
9. Target Value	File type/size validated; malware scan applied before storage
10. Priority	Critical
11. Risk	High
12. Dependencies	XSS Protection (NFR-SEC-002)
13. Applicable Modules	Admission & Enrollment
14. Related FRs (Appendix-E)	FR-01
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	File-type allowlisting, size limits, malware scanning
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic malicious-file-upload penetration testing
18. Validation Method	Security code review
19. Verification Method	Malicious-file-upload penetration testing
20. Compliance Standard	OWASP Top 10
21. Remarks	No outstanding client input required for this NFR.

Category: Network Security (2 NFRs)**NFR-NET-001 — TLS/HTTPS Enforcement**

1. NFR ID	NFR-NET-001
2. Title	TLS/HTTPS Enforcement
3. Category	Network Security
4. Description	All traffic is served exclusively over HTTPS with TLS 1.2+; HTTP requests are redirected.
5. Business Objective	Protect all data in transit, including login credentials and payment references.
6. Rationale	Foundational network security control; a prerequisite for nearly every other security NFR to be meaningful.
7. Acceptance Criteria	SSL/TLS configuration scan confirms no unencrypted endpoint is reachable.
8. Measurement Method	Network/SSL configuration scan
9. Target Value	100% of traffic over HTTPS, TLS 1.2+, HTTP redirected
10. Priority	Critical
11. Risk	High
12. Dependencies	Encryption in Transit (NFR-ENC-002)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	HTTPS enforcement, HSTS, TLS 1.2+
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic network/ssl configuration scan (e.g., ssl labs-equivalent)
18. Validation Method	Security code review
19. Verification Method	Network/SSL configuration scan (e.g., SSL Labs-equivalent)
20. Compliance Standard	OWASP ASVS
21. Remarks	No outstanding client input required for this NFR.

NFR-NET-002 — API/Token Security

1. NFR ID	NFR-NET-002
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2. Title	API/Token Security
3. Category	Network Security
4. Description	API authentication uses short-lived, signed tokens (JWT or equivalent) with a defined refresh mechanism.
5. Business Objective	Protect API endpoints, including third-party integrations (Payment Gateway, SMS Gateway), from unauthorized access.
6. Rationale	Directly relevant to the External Integrations already flagged in RGD v2.0 Section 15.
7. Acceptance Criteria	Penetration test confirms tokens expire correctly and cannot be replayed or forged.
8. Measurement Method	Penetration testing of API authentication flow
9. Target Value	Short-lived signed tokens (e.g., JWT) with refresh mechanism
10. Priority	Critical
11. Risk	High
12. Dependencies	TLS/HTTPS Enforcement (NFR-NET-001)
13. Applicable Modules	Fee Collection, Communication & Notifications, Transport Management
14. Related FRs (Appendix-E)	FR-24, FR-37, FR-31
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	JWT (or equivalent) with signature verification and expiry
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic network/ssl configuration scan (e.g., ssl labs-equivalent)
18. Validation Method	Security code review
19. Verification Method	Network/SSL configuration scan (e.g., SSL Labs-equivalent)
20. Compliance Standard	OWASP ASVS
21. Remarks	No outstanding client input required for this NFR.

Category: Monitoring (2 NFRs)**NFR-MON-001 — System Health Monitoring**

1. NFR ID	NFR-MON-001
2. Title	System Health Monitoring
3. Category	Monitoring
4. Description	Real-time monitoring of uptime, response time, and error rate, visible to IT Admin.
5. Business Objective	Enable proactive issue detection before it becomes a user-facing outage.
6. Rationale	Directly supports the Availability (NFR-AVAIL-001) and Reliability (NFR-REL-001) NFRs by providing the underlying data.
7. Acceptance Criteria	Monitoring dashboard is live and demonstrably reflects real system state during a simulated incident.
8. Measurement Method	Monitoring-dashboard review and alert-trigger test
9. Target Value	Real-time visibility into uptime, latency, and error rate
10. Priority	High
11. Risk	Medium
12. Dependencies	Alerting (NFR-ALT-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic monitoring-dashboard review and alert-trigger test
18. Validation Method	Monitoring configuration audit
19. Verification Method	Monitoring-dashboard review and alert-trigger test
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

NFR-MON-002 — Third-Party Integration Monitoring

1. NFR ID	NFR-MON-002
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2. Title	Third-Party Integration Monitoring
3. Category	Monitoring
4. Description	Independent health monitoring for each external integration (Payment Gateway, SMS/Email Gateway, GPS, UIDAI).
5. Business Objective	Detect integration-specific failures separately from core system health, given FR-37's cross-module single-point-of-failure risk (Appendix-D).
6. Rationale	Directly addresses the Appendix-D Dependency Risk Register entry for FR-37's shared notification service.
7. Acceptance Criteria	Each integration has an independent health check with alerting on failure.
8. Measurement Method	Monitoring-dashboard review per integration
9. Target Value	Independent health status per integration, updated at a defined interval
10. Priority	High
11. Risk	High
12. Dependencies	System Health Monitoring (NFR-MON-001), Alerting (NFR-ALT-001)
13. Applicable Modules	Fee Collection, Communication & Notifications, Transport Management, Admission & Enrollment
14. Related FRs (Appendix-E)	FR-24, FR-37, FR-31, FR-05a
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic monitoring-dashboard review and alert-trigger test
18. Validation Method	Monitoring configuration audit
19. Verification Method	Monitoring-dashboard review and alert-trigger test
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Alerting (1 NFR)**NFR-ALT-001 — Automated Critical-Failure Alerting**

1. NFR ID	NFR-ALT-001
2. Title	Automated Critical-Failure Alerting
3. Category	Alerting
4. Description	Critical failures (payment gateway down, backup failure, unusual login pattern) trigger automated alerts to IT Admin.
5. Business Objective	Minimize detection-to-response time for high-impact failures.
6. Rationale	Monitoring without alerting only informs someone who happens to be watching; alerting ensures timely human response.
7. Acceptance Criteria	Alert-trigger simulation confirms IT Admin receives the alert within target time.
8. Measurement Method	Alert-trigger simulation test
9. Target Value	Alert delivered within 5 minutes of trigger condition
10. Priority	Critical
11. Risk	High
12. Dependencies	Monitoring series
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic alert-trigger simulation test
18. Validation Method	Alerting configuration audit
19. Verification Method	Alert-trigger simulation test
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Error Handling (1 NFR)**NFR-ERR-001 — Consistent Error Handling Pattern**

1. NFR ID	NFR-ERR-001
2. Title	Consistent Error Handling Pattern
3. Category	Error Handling
4. Description	All modules follow the standardized error-handling pattern: inline validation errors, logged system errors with user-friendly messaging, no silent data loss.
5. Business Objective	Ensure a predictable, trustworthy user experience and complete diagnosability for IT Admin.
6. Rationale	Directly operationalizes the Error Handling pattern already defined once and referenced by every requirement in Appendix-E Field 37.
7. Acceptance Criteria	Negative-path test suite confirms every tested failure mode surfaces an appropriate message and is logged, with no silent failure.
8. Measurement Method	Negative-path test suite execution
9. Target Value	100% of tested failure modes handled per the standard pattern
10. Priority	High
11. Risk	Medium
12. Dependencies	Logging series, Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic negative-path test suite execution
18. Validation Method	Cross-reference against Appendix-E Field 37
19. Verification Method	Negative-path test suite execution
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Logging (10 NFRs)**NFR-LOG-001 — Authentication Logging**

1. NFR ID	NFR-LOG-001
2. Title	Authentication Logging
3. Category	Logging
4. Description	All login attempts, successful and failed, are logged with timestamp, user, and IP address.
5. Business Objective	Support security investigation of suspicious access patterns.
6. Rationale	A foundational security-logging control, feeding NFR-MON-001's "unusual login pattern" alerting.
7. Acceptance Criteria	Sampling audit confirms 100% of login attempts (success and failure) are logged with required fields.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of login attempts logged
10. Priority	Critical
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-002 — Fee Transaction Logging

1. NFR ID	NFR-LOG-002
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2. Title	Fee Transaction Logging
3. Category	Logging
4. Description	All payment, refund, and credit-note actions are logged with full before/after detail.
5. Business Objective	Support financial audit and dispute resolution.
6. Rationale	Directly supports enforcement visibility for BR-FEE-001 (paid-invoice immutability) and BR-FEE-002 (refund authority).
7. Acceptance Criteria	Sampling audit confirms 100% of tested fee transactions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of fee transactions logged
10. Priority	Critical
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Fee Collection
14. Related FRs (Appendix-E)	FR-23, FR-24, FR-25, FR-26
15. Related Business Rules (Appendix-C)	BR-FEE-001, BR-FEE-002
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-003 — Attendance Logging

1. NFR ID	NFR-LOG-003
2. Title	Attendance Logging
3. Category	Logging
4. Description	All attendance marks and corrections are logged, including the correction-workflow audit trail.
5. Business Objective	Support dispute resolution and payroll/examination-eligibility audit.

6. Rationale	Directly supports enforcement visibility for BR-ATT-001 and BR-ATT-002.
7. Acceptance Criteria	Sampling audit confirms 100% of tested attendance actions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of attendance actions logged
10. Priority	High
11. Risk	Medium
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Attendance Management
14. Related FRs (Appendix-E)	FR-10
15. Related Business Rules (Appendix-C)	BR-ATT-001, BR-ATT-002
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-004 — Exam Logging

1. NFR ID	NFR-LOG-004
2. Title	Exam Logging
3. Category	Logging
4. Description	All marks entry, subject locks, and re-evaluation overrides are logged with before/after values.
5. Business Objective	Preserve academic-integrity audit trail.
6. Rationale	Directly supports enforcement visibility for BR-EXM-003 (locked marks immutability).
7. Acceptance Criteria	Sampling audit confirms 100% of tested exam actions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of exam actions logged

10. Priority	Critical
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Examination & Result Processing
14. Related FRs (Appendix-E)	FR-18, FR-19
15. Related Business Rules (Appendix-C)	BR-EXM-002, BR-EXM-003
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-005 — Library Logging

1. NFR ID	NFR-LOG-005
2. Title	Library Logging
3. Category	Logging
4. Description	All issue, return, and fine transactions are logged.
5. Business Objective	Support catalog-integrity and fine-dispute resolution.
6. Rationale	Lower sensitivity than financial/academic logs but still required for consistent module coverage.
7. Acceptance Criteria	Sampling audit confirms 100% of tested library transactions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of library transactions logged
10. Priority	Medium
11. Risk	Low
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Library Management

14. Related FRs (Appendix-E)	FR-28
15. Related Business Rules (Appendix-C)	BR-LIB-002, BR-LIB-003
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-006 — Transport Logging

1. NFR ID	NFR-LOG-006
2. Title	Transport Logging
3. Category	Logging
4. Description	All allocation changes and trip logs are retained.
5. Business Objective	Support safety-audit requirements for student transportation.
6. Rationale	Trip logs are explicitly required per Appendix-A Module 7 post-conditions for safety audit.
7. Acceptance Criteria	Sampling audit confirms 100% of tested transport actions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of transport allocation changes and trips logged
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Transport Management
14. Related FRs (Appendix-E)	FR-30, FR-31
15. Related Business Rules (Appendix-C)	BR-TRN-001, BR-TRN-004
16. Related Security Controls	N/A

17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-007 — HR Logging

1. NFR ID	NFR-LOG-007
2. Title	HR Logging
3. Category	Logging
4. Description	All staff master changes and access-revocation events are logged.
5. Business Objective	Support security and employment-record audit.
6. Rationale	Directly supports enforcement visibility for BR-HR-002 (access revocation on exit) — a security-critical event.
7. Acceptance Criteria	Sampling audit confirms 100% of tested HR actions, especially access revocation, produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of staff master changes and access events logged
10. Priority	Critical
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	HR & Payroll
14. Related FRs (Appendix-E)	FR-33
15. Related Business Rules (Appendix-C)	BR-HR-002, BR-HR-006
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4

21. Remarks	No outstanding client input required for this NFR.
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NFR-LOG-008 — Payroll Logging

1. NFR ID	NFR-LOG-008
2. Title	Payroll Logging
3. Category	Logging
4. Description	All payroll runs, approvals, and adjustments are logged.
5. Business Objective	Support statutory and financial audit of compensation processing.
6. Rationale	Directly supports enforcement visibility for BR-HR-003 (no duplicate payroll run).
7. Acceptance Criteria	Sampling audit confirms 100% of tested payroll actions produce a complete log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of payroll runs and adjustments logged
10. Priority	Critical
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	HR & Payroll
14. Related FRs (Appendix-E)	FR-34, FR-36
15. Related Business Rules (Appendix-C)	BR-HR-001, BR-HR-003, BR-HR-007
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-009 — Configuration Change Logging

1. NFR ID	NFR-LOG-009
2. Title	Configuration Change Logging
3. Category	Logging
4. Description	All changes to fee structure, grading scheme, and system configuration are logged with before/after values.
5. Business Objective	Prevent silent, unauthorized configuration drift affecting many downstream transactions.
6. Rationale	A single configuration change (e.g., fee structure) can affect thousands of subsequent invoices; must be fully traceable.
7. Acceptance Criteria	Sampling audit confirms 100% of tested configuration changes produce a complete before/after log entry.
8. Measurement Method	Log-completeness sampling audit
9. Target Value	100% of configuration changes logged
10. Priority	High
11. Risk	High
12. Dependencies	Audit Trail (NFR-AUDIT-001)
13. Applicable Modules	Fee Collection, Examination, All configuration-bearing modules
14. Related FRs (Appendix-E)	FR-22, FR-17
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

NFR-LOG-010 — Consolidated Audit Trail Access

1. NFR ID	NFR-LOG-010
2. Title	Consolidated Audit Trail Access
3. Category	Logging
4. Description	All logs above are queryable centrally by IT Admin for audit and investigation purposes.

5. Business Objective	Avoid a fragmented, module-siloed logging landscape that impedes investigation.
6. Rationale	Without central queryability, the individual per-module logs above have limited practical audit value.
7. Acceptance Criteria	IT Admin can query across all module logs from a single interface, filtered by user, date range, and action type.
8. Measurement Method	Centralized log-query interface test
9. Target Value	Single-pane query across all module logs
10. Priority	High
11. Risk	Medium
12. Dependencies	All Logging NFRs above
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Cross-reference against Appendix-E Field 24 and module BRULEs
19. Verification Method	Log-completeness sampling audit
20. Compliance Standard	ISO 27001 Annex A.12.4
21. Remarks	No outstanding client input required for this NFR.

Category: Maintainability (1 NFR)

NFR-MAINT-001 — Code Maintainability Standard

1. NFR ID	NFR-MAINT-001
2. Title	Code Maintainability Standard
3. Category	Maintainability
4. Description	Codebase follows documented coding standards and modular architecture enabling isolated module updates.
5. Business Objective	Reduce long-term cost of ownership and risk of regression when modifying one module.
6. Rationale	Given 11 modules with significant cross-dependencies (Appendix-D), poor modularity would make maintenance increasingly costly over time.
7. Acceptance Criteria	Static-analysis metrics (complexity, coupling) remain within agreed thresholds; architecture review confirms module boundaries are respected.
8. Measurement Method	Code-quality static analysis
9. Target Value	<i>Coupling/complexity metrics within agreed thresholds (Client/Product Decision Required for specific tool/thresholds)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Extensibility (NFR-EXT-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Architecture/code review
19. Verification Method	Code-quality static analysis (complexity, coupling metrics)
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Extensibility (1 NFR)**NFR-EXT-001 — Modular Extensibility**

1. NFR ID	NFR-EXT-001
2. Title	Modular Extensibility
3. Category	Extensibility
4. Description	Architecture supports adding new modules (e.g., future Alumni or Hostel management) without re-architecting existing modules.
5. Business Objective	Protect the client's investment as institutional needs grow beyond the current 11-module scope.
6. Rationale	RGD v2.0 Section 2.1 scope is fixed for this phase, but the architecture should not foreclose reasonable future growth.
7. Acceptance Criteria	Architecture review confirms a documented extension pattern (e.g., plugin/module registration approach) exists.
8. Measurement Method	Architecture review against a sample future-module addition
9. Target Value	Documented, demonstrable module-extension pattern
10. Priority	Medium
11. Risk	Low
12. Dependencies	Maintainability (NFR-MAINT-001), Configurability (NFR-CFG-001)
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	N/A
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Design review
19. Verification Method	Architecture review against a sample future-module addition
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Configurability (1 NFR)**NFR-CFG-001 — Business Rule Configurability**

1. NFR ID	NFR-CFG-001
2. Title	Business Rule Configurability
3. Category	Configurability
4. Description	Threshold/rate/policy values flagged across Appendix-C Section 3.5 must be configurable by IT Admin without code changes.
5. Business Objective	Avoid a costly code-change cycle every time the school adjusts a policy value (e.g., late-fee rate, attendance threshold).
6. Rationale	Directly operationalizes the 24 configurable items already catalogued in Appendix-C v1.1 Section 3.5.
7. Acceptance Criteria	Configuration test confirms each of the 24 Appendix-C configurable items can be changed via an admin interface without a code deployment.
8. Measurement Method	Configuration-change test without code deployment
9. Target Value	100% of Appendix-C Section 3.5 items configurable via UI/admin interface
10. Priority	High
11. Risk	Medium
12. Dependencies	N/A
13. Applicable Modules	All Modules
14. Related FRs (Appendix-E)	All FRs
15. Related Business Rules (Appendix-C)	All rules referenced in Appendix-C Section 3.5
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic configuration-change test without code deployment
18. Validation Method	Cross-reference against Appendix-C Section 3.5 configurable items
19. Verification Method	Configuration-change test without code deployment
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Localization (1 NFR)

NFR-L10N-001 — Regional Language Support

1. NFR ID	NFR-L10N-001
2. Title	Regional Language Support
3. Category	Localization
4. Description	UI and report card templates support Hindi and one additional regional language alongside English.
5. Business Objective	Meet the localization expectation set in RGD v2.0 Section 19 for the Indian school context.
6. Rationale	Explicitly required per RGD v2.0 Non-Functional Requirements (Localization).
7. Acceptance Criteria	Language-toggle test confirms UI and report cards render correctly in all three configured languages.
8. Measurement Method	Language-toggle functional testing plus native-speaker review
9. Target Value	<i>English, Hindi, and one additional regional language (Client/Product Decision Required to confirm which)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Internationalization (NFR-I18N-001)
13. Applicable Modules	All Modules (UI-wide), Examination (report cards)
14. Related FRs (Appendix-E)	FR-20
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic language-toggle functional testing
18. Validation Method	Native-speaker content review
19. Verification Method	Language-toggle functional testing
20. Compliance Standard	RGD v2.0 NFR (Localization)
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Internationalization (1 NFR)**NFR-I18N-001 — Currency/Date Format Framework**

1. NFR ID	NFR-I18N-001
2. Title	Currency/Date Format Framework
3. Category	Internationalization
4. Description	System architecture supports multiple currency/date formats even though INR/DD-MM-YYYY is the only active configuration at go-live.
5. Business Objective	Avoid a hard-coded, India-only assumption that would block future multi-region use if the client expands.
6. Rationale	A forward-looking architecture decision distinct from the immediate localization need (NFR-L10N-001).
7. Acceptance Criteria	Architecture review confirms currency/date formatting is abstracted, not hard-coded, even with only one active locale.
8. Measurement Method	Architecture review
9. Target Value	Locale-abstracted currency/date formatting framework
10. Priority	Low
11. Risk	Low
12. Dependencies	Regional Language Support (NFR-L10N-001)
13. Applicable Modules	Fee Collection, All Modules (date fields)
14. Related FRs (Appendix-E)	FR-23, FR-26
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic currency/date-format framework test with a secondary locale
18. Validation Method	Architecture review
19. Verification Method	Currency/date-format framework test with a secondary locale
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Data Retention (1 NFR)

NFR-RET-001 — Retention Policy Enforcement

1. NFR ID	NFR-RET-001
2. Title	Retention Policy Enforcement
3. Category	Data Retention
4. Description	Student, staff, and financial records are retained per a configured policy before archival eligibility.
5. Business Objective	Balance statutory retention obligations against storage cost and Data Privacy minimization principles.
6. Rationale	DPDP Act, 2023 requires data not be retained longer than necessary, creating tension with statutory record-keeping obligations that must be resolved by policy, not left ambiguous.
7. Acceptance Criteria	Retention-policy test confirms records become archival-eligible exactly at the configured threshold, not before or after.
8. Measurement Method	Retention-policy enforcement test
9. Target Value	<i>Policy period Client/Product Decision Required (to be confirmed with client's statutory/legal guidance)</i>
10. Priority	High
11. Risk	Medium
12. Dependencies	Archiving (NFR-ARC-001), Backup Retention (NFR-BKP-002)
13. Applicable Modules	Student Information Management, HR & Payroll, Fee Collection
14. Related FRs (Appendix-E)	FR-08, FR-33, FR-23
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic retention-policy enforcement test (data eligible for archival is correctly identified)
18. Validation Method	Policy-configuration audit
19. Verification Method	Retention-policy enforcement test (data eligible for archival is correctly identified)
20. Compliance Standard	DPDP Act, 2023
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Archiving (1 NFR)

NFR-ARC-001 — Historical Data Archiving

1. NFR ID	NFR-ARC-001
2. Title	Historical Data Archiving
3. Category	Archiving
4. Description	Closed academic-year data is archived to lower-cost storage after a configured period while remaining queryable.
5. Business Objective	Control active-database growth (see Scalability series) while preserving historical accessibility.
6. Rationale	Directly extends BR-SIS-004 (Historical Record Immutability) with a storage-tiering strategy.
7. Acceptance Criteria	Archive-and-retrieve test confirms archived data remains fully queryable within an acceptable retrieval time.
8. Measurement Method	Archive-and-retrieve test cycle
9. Target Value	<i>Closed-year data archived after Client/Product Decision Required, remains queryable within an acceptable retrieval time (Client/Product Decision Required)</i>
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Data Retention (NFR-RET-001), Database & Storage Growth (NFR-SCAL-008)
13. Applicable Modules	Student Information Management, Examination
14. Related FRs (Appendix-E)	FR-08
15. Related Business Rules (Appendix-C)	BR-SIS-004
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic archive-and-retrieve test cycle
18. Validation Method	Archive-integrity spot check
19. Verification Method	Archive-and-retrieve test cycle
20. Compliance Standard	Internal Quality Standard
21. Remarks	Contains one or more values pending client/product confirmation — see flagged fields above.

Category: Compliance (1 NFR)**NFR-COMP-001 — Regulatory Compliance Adherence**

1. NFR ID	NFR-COMP-001
2. Title	Regulatory Compliance Adherence
3. Category	Compliance
4. Description	System design demonstrably satisfies every item in RGD v2.0 Section 16 (Regulatory Compliance).
5. Business Objective	Ensure the delivered system does not create statutory or contractual exposure for the school.
6. Rationale	Consolidates RTE Act 2009, DPDP Act 2023, UDISE+, Board bye-laws, GST, and Shops & Establishment Act obligations into a single trackable NFR.
7. Acceptance Criteria	Compliance checklist confirms each RGD v2.0 Section 16 item has a corresponding, implemented system control.
8. Measurement Method	Compliance checklist audit against RGD v2.0 Section 16
9. Target Value	100% of RGD v2.0 Section 16 items addressed with a traceable system control
10. Priority	Critical
11. Risk	High
12. Dependencies	Privacy (NFR-PRIV-001), all Regulatory-flagged Business Rules
13. Applicable Modules	Admission & Enrollment, Fee Collection, HR & Payroll
14. Related FRs (Appendix-E)	FR-05a, FR-05b, FR-05c, FR-26a, FR-26b, FR-34
15. Related Business Rules (Appendix-C)	BR-ADM-003, BR-ADM-004, BR-FEE-005, BR-FEE-007, BR-HR-005
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Legal/compliance review
19. Verification Method	Compliance checklist audit against RGD v2.0 Section 16
20. Compliance Standard	RTE Act 2009; DPDP Act 2023; UDISE+; GST; Shops & Establishment Act
21. Remarks	No outstanding client input required for this NFR.

Category: Privacy (1 NFR)**NFR-PRIV-001 — Data Privacy by Design**

1. NFR ID	NFR-PRIV-001
2. Title	Data Privacy by Design
3. Category	Privacy
4. Description	Personal data collection is limited to necessity (data minimization) with documented consent capture.
5. Business Objective	Comply with DPDP Act, 2023 principles and build institutional trust with parents/staff.
6. Rationale	Especially significant given the volume of minors' personal data (students) processed by this system.
7. Acceptance Criteria	Privacy impact assessment confirms no personal data field is collected without a documented purpose and consent basis.
8. Measurement Method	Privacy impact assessment
9. Target Value	100% of collected personal data fields have a documented purpose and consent basis
10. Priority	Critical
11. Risk	High
12. Dependencies	Encryption at Rest (NFR-ENC-001), Data Retention (NFR-RET-001)
13. Applicable Modules	Admission & Enrollment, Student Information Management, HR & Payroll
14. Related FRs (Appendix-E)	FR-01, FR-05a, FR-06, FR-33
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	Consent management, data minimization by design
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002
18. Validation Method	Privacy impact assessment
19. Verification Method	Data-minimization and consent-flow audit
20. Compliance Standard	DPDP Act, 2023
21. Remarks	No outstanding client input required for this NFR.

Category: Reporting Performance (1 NFR)**NFR-RPTP-001 — Large Report Background Processing**

1. NFR ID	NFR-RPTP-001
2. Title	Large Report Background Processing
3. Category	Reporting Performance
4. Description	Reports exceeding a row-count threshold process as background jobs with progress indication, not a blocking UI call.
5. Business Objective	Prevent large report requests from degrading the experience of all concurrent users.
6. Rationale	Directly extends the Report Generation (NFR-PERF-005) and Export (NFR-PERF-009) targets for the large-report edge case.
7. Acceptance Criteria	Load test with a maximum-scale report confirms the UI remains responsive and the user is notified on completion.
8. Measurement Method	Large-report load testing
9. Target Value	Reports >10,000 rows processed as background jobs with completion notification
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Report Generation (NFR-PERF-005), Export (NFR-PERF-009)
13. Applicable Modules	Reports & Analytics, Fee Collection
14. Related FRs (Appendix-E)	FR-26, FR-41, FR-42
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic large-report load testing
18. Validation Method	Background-job queue test
19. Verification Method	Large-report load testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Search Performance (1 NFR)**NFR-SRCH-001 — Indexed Search at Scale**

1. NFR ID	NFR-SRCH-001
2. Title	Indexed Search at Scale
3. Category	Search Performance
4. Description	Student/staff/catalog search is backed by indexed search infrastructure, maintaining target response time even at maximum scale.
5. Business Objective	Ensure search remains usable as the student/staff/catalog base grows toward Scalability targets.
6. Rationale	Directly extends Search Response (NFR-PERF-004) to explicitly cover the maximum-scale scenario, not just current-day volume.
7. Acceptance Criteria	Load test at maximum Scalability-target data volume confirms search response remains within the Performance NFR target.
8. Measurement Method	Search-index load testing at maximum scale
9. Target Value	≤2 seconds at 20,000-student / 30,000-guardian scale (per NFR-SCAL-002/004)
10. Priority	Medium
11. Risk	Medium
12. Dependencies	Search Response (NFR-PERF-004), Maximum Students (NFR-SCAL-002)
13. Applicable Modules	Student Information Management, Library Management
14. Related FRs (Appendix-E)	FR-06, FR-27, FR-29
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic search-index load testing at maximum scale
18. Validation Method	Search relevance/performance test suite
19. Verification Method	Search-index load testing at maximum scale
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Category: Print Performance (1 NFR)**NFR-PRN-001 — Print Rendering Time**

1. NFR ID	NFR-PRN-001
2. Title	Print Rendering Time
3. Category	Print Performance
4. Description	ID cards, report cards, and receipts render print-ready within a target time, individually and in batch.
5. Business Objective	Support time-sensitive bulk-printing operations (e.g., ID cards for an entire incoming class).
6. Rationale	Directly relevant to FR-09 (ID Card generation) and FR-20 (Report Card generation), both of which are typically run in batch.
7. Acceptance Criteria	Batch-print load test confirms a full-class batch completes within target.
8. Measurement Method	Batch-print load testing
9. Target Value	≤3 seconds per document; ≤5 minutes for a 30-40 document class batch
10. Priority	Low
11. Risk	Low
12. Dependencies	Report Generation (NFR-PERF-005)
13. Applicable Modules	Student Information Management, Examination, Fee Collection
14. Related FRs (Appendix-E)	FR-09, FR-20, FR-23
15. Related Business Rules (Appendix-C)	N/A
16. Related Security Controls	N/A
17. Monitoring Method	Continuous automated monitoring per NFR-MON-001/002, supplemented by periodic batch-print load testing
18. Validation Method	Print-rendering test suite
19. Verification Method	Batch-print load testing
20. Compliance Standard	Internal Quality Standard
21. Remarks	No outstanding client input required for this NFR.

Quality Validation

Programmatically verified at build time against the underlying NFR dataset.

Total NFRs

88 Non-Functional Requirements defined.

Coverage by Category

Category	NFR Count	Status
Performance	12	Covered
Scalability	8	Covered
Availability	2	Covered
Reliability	2	Covered
Security	3	Covered
Authentication	2	Covered
Authorization	1	Covered
Audit Logging	1	Covered
Data Integrity	2	Covered
Backup	5	Covered
Disaster Recovery	2	Covered
Business Continuity	1	Covered
Usability	2	Covered
Accessibility	5	Covered
Browser Compatibility	1	Covered
Mobile Compatibility	1	Covered
API Performance	1	Covered
Database Performance	1	Covered
Caching	1	Covered

Session Management	1	Covered
Password Policy	3	Covered
Encryption	2	Covered
File Upload Security	1	Covered
Network Security	2	Covered
Monitoring	2	Covered
Alerting	1	Covered
Error Handling	1	Covered
Logging	10	Covered
Maintainability	1	Covered
Extensibility	1	Covered
Configurability	1	Covered
Localization	1	Covered
Internationalization	1	Covered
Data Retention	1	Covered
Archiving	1	Covered
Compliance	1	Covered
Privacy	1	Covered
Reporting Performance	1	Covered
Search Performance	1	Covered
Print Performance	1	Covered

Coverage by Module

Applicable Modules referenced across all NFRs: Admission & Enrollment, All Modules (UI-wide), All Modules (date fields), All configuration-bearing modules, Attendance, Attendance Management, Communication, Communication & Notifications, Examination, Examination & Result Processing, Examination (report cards), Fee Collection, HR, HR & Payroll, Library, Library Management, Parent/Student Portal, Parent/Student Portal (all modules with portal exposure), Reports & Analytics, SIS, Student Information Management, Timetable & Scheduling, Transport Management. In addition, 44 NFRs are explicitly cross-cutting ("All Modules").

Coverage by Functional Requirement (Appendix-E)

42 distinct Functional Requirement IDs are explicitly referenced by at least one NFR (in addition to NFRs marked "All FRs", which apply system-wide across all 47).

Coverage by Business Rule (Appendix-C)

32 distinct Business Rule IDs are explicitly referenced by at least one NFR, concentrated in the immutability, authorization, and regulatory rule families.

Missing Items

None — all 40 requested NFR categories have at least one defined NFR.

No duplicate NFR IDs: confirmed (88 unique IDs across 88 records).

Quality Score

92/100 — deductions reflect the number of target values still pending client confirmation (marked Client/Product Decision Required) rather than any structural gap in coverage or traceability.

Approval Status

Conditionally Ready — all 40 categories are covered with zero missing items and zero duplicate IDs; approval is gated on the client confirming the specific numeric targets and policy values flagged Client/Product Decision Required throughout this catalogue (concurrency figures, statutory slabs, retention periods, SLA windows), consistent with the same category of open items already tracked in Appendix-C v1.1 and Appendix-D.